

# Harmonic oscillator 1

Let  $V(x)$  be the potential energy for a harmonic oscillator.

$$V(x) = \frac{m\omega^2 x^2}{2}$$

We seek  $\psi(x, t)$  that solves the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t)$$

The solution is

$$\psi_n(x, t) = C_n \exp \left[ \frac{m\omega x^2}{2\hbar} - i\omega t \left( n + \frac{1}{2} \right) \right] \frac{d^n}{dx^n} \exp \left( -\frac{m\omega x^2}{\hbar} \right)$$

where  $n = 0, 1, 2, \dots$  and  $C_n$  is the normalization constant

$$C_n = \frac{(-1)^n}{\sqrt{2^n n!}} \left( \frac{m\omega}{\pi \hbar} \right)^{1/4} \left( \frac{\hbar}{m\omega} \right)^{n/2}$$

By inspection of the time exponential in  $\psi_n(x, t)$  we find that

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \hbar\omega \left( n + \frac{1}{2} \right) \psi(x, t)$$

Hence  $\psi(x, t)$  also solves the time-independent Schrodinger equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t) = E_n \psi(x, t) \tag{1}$$

where for the harmonic oscillator

$$E_n = \hbar\omega \left( n + \frac{1}{2} \right)$$

The time exponential of  $\psi_n(x, t)$  cancels in (1) hence  $\psi_n(x)$  is also a solution where

$$\psi_n(x) = C_n \exp \left( \frac{m\omega x^2}{2\hbar} \right) \frac{d^n}{dx^n} \exp \left( -\frac{m\omega x^2}{\hbar} \right)$$

Eigenmath script